

Sagar Vekariya



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(+091) 81 55 80 77 13 | Surat – permanent | Ahmedabad – current | ready to relocate

B. Tech in Mechanical | PGDM in Operations Management

Many Technologies Courses | CAD | CAE | CFD | SolidWorks etc.

Problem-Solving | Data-driven decision | Businesses | Initiator | Calculated risk taker

CAREER OBJECTIVE

To secure a position where I can effectively apply my problem-solving skills, creative thinking, and design thinking abilities to foster a stress-free work environment. Aiming to lead and build successful teams through collaborative efforts while pursuing continuous learning and growth.

ACADEMIC QUALIFICATIONS

- 10th passed in 2012 from Sadhana Vidyalaya with 83 % percentile (GSEB board)
- 12th Science passed in 2014 from Jay Ambe International with 92% percentile (GSHEB board)
- Completed B. Tech in Mechanical Engineering (in 2014-2018) from ITMVU with an 8.87 CGPA
- Completed PGDM in Operation Management (in 2019-2022) from NMIMS Global with 56 %

TECHNICAL QUALIFICATION

- **Diploma in Product Design & Analysis from CADD Canter** 2016-2018
 - 2D Drafting & 3D Moulding (CAD)
 - FEA & Basic CFD using Ansys Workbench
- **Skill – Lync (Pursuing)** 2021- 2023
 - Post Graduate Program in Computational Fluid Dynamics
 - Post Graduate Program in Hybrid Electric Vehicle Design and Analysis
- **ROBOAI: An Industrial Training Program** 2024 (3 month)
Robotics - Mechanical & Industrial Design
Electronics & IOT
ML/AI ROS-2

Software – Hands-on Experience

- **Designing Tools:** - Solid Works (2014 to 2024), Ansys (2016 to 2024), Cura (FDM 3D Slicer), CHITUBOX & Other 3D Slicers of 3D Printers, Autodesk Netfabb, Autodesk Meshmixer, Matlab, Minitab, Auto Cad (Rarely used after 2019), Knowledge and use 5% when required like (Creo, CATIA, Siemens Edge, GT Power, Ansa)
- **Microsoft Office 365, Basic Windows operating and troubleshooting, etc.**

Current Experience

Harikrishna Technopride engg., Ahmedabad, Gujarat.

Apil 2024 – till date

Design Manager

Harikrushna Technopride Engg. is one of the leading manufacturers & exporters of various kinds of pharmaceutical machinery. We are specialists In injectable, **Prefilled Syringe Filling And Stoppering (Closing) Machine, Pre Filled Syringe Filling Machine.**

We manufacture injectable & packaging machinery like washing, filling, sealing, inspection, Labeling and conveyers. Harikrushna Technopride Engg. was established in 1996. Our plant at Vatva (G.I.D.C.) in Ahmedabad, Gujarat (India), is equipped with the latest production and quality testing facilities. Our team consists of qualified engineers with vast experience in the manufacture **of pharmaceuticals and packaging machinery**.

Key Performance Indicators (KPIs)

- Leading the development of pharmaceutical packaging machinery from just idea or customer requirement and release of system/electro-mechanical modules or re-design existing products to improve overall long-term system quality, reliability, yield rates, and cost effectiveness.
- Engineering Change Order (ECO) process maintain for upgrade existing design/drawings and other related documents
- Working with a cross-functional team to ensure the design and development of electro-mechanical components to meet the needs of the internal and external stockholders or Customers.
- Using modelling and simulation tools to create and modify 3D designs, and 2D drawings.
- Drive continuous improvement for ongoing projects.
- Fulfil the all-type requirements of internal and external customer.
- Ensuring that the design is consistent with the higher-level architecture and requirements
- Responsible for ensuring that the design meets the performance, quality and cost criteria
- Ensuring that there is proper documentation per standards for the developed design
- Ensuring the design that meets the product specification
- Team building and mentoring the junior members.

Responsibilities

- Manage, insert and maintain the product data codifications into the company tools and systems.
- Participate and Lead the Project meetings with clients.
- Support clients during assembly and testing of the designed components, to solve any issue which can occur during these activities.
- Prepare technical documentation of their deliverables.

Tool: - SolidWorks

Team Size: - 3

Past Experience

TK Agro Pvt. Ltd., Indore, Madhya Pradesh.

July 2023 – March 2024

R&D Head.

Responsible for Developing **Innovative Farming equipment** like, for **Soil Preparation** (Ploughing, cultivator, Rotavator, etc), **Corp Plantings, & Corp harvesting**.

Project: -1 Garlic & Onion Planter Development.

Project: -2 Reaper Binder – **(NPD Recently completed)**

Project: -3 Seed drill – **NPD** (wheat, Corn, Grams, jwar, Bazara(millet), Peanuts, Soybean, sunflower, etc).

project: -4 Plough

project: -5 Rotavator

Managing Design engineers' team.

Documentation and system development for the R&D department.

Preparing a Manufacturing Operations procedure.

Designed a customized Jig and fixture to reduce human efforts, and manufacturing time saving.

Designed die components for sheet metal forming components.

Creating and maintaining FMEA data.

Used **different problem-solving techniques** to improve product quality.

Skills used: - CAD (Solid works, AutoCAD), CAE (Ansys, Solid works), Engineering Drawing, Motion Simulations, Reverse Engineering, 3D Printing, Fabrication and sheet metal, Casting, forming Die developing, People management, Cross-function team coordination, etc.

Team Sized: - 5 people teams.

US Institute of 3D Technology India Pt. Ltd., Ahmedabad, Gujarat

Oct 2020 – Jun 2023

CTO and Manager

USI3DT based in Orinda, California, USA, who **introduced first-time 3D Printing Technology in Gujarat state and India's central government** also **helped to central government to develop a policy for additive manufacturing**. In this organization, I am **responsible for driving innovation, ensuring product quality and performance**, managing a talented team, and maintaining compliance with industry standards while aligning **technology strategy** with the **business goals and objectives of the company**. Signed MOU with **36+** colleges for **Canter of Excellence** of additive manufacturing. **Developed** a completely **new business** in the **STEM Education** filed and **built a brand** known as **US STEM CLUB**. **Build a new business in healthcare** known as Lak Techne India Pvt. Ltd. Developed **Technical Strategy for 3D Technology in the Healthcare Industry** (Surgery Planning – Visualization - 3D Printing, Implant Design).

PRG Agitators Pvt Ltd

March 2018 - Sept 2020

Design and Development Engineer

It is a German company. I give my service here as a **design and development engineer** in the design and development department at PRG Agitators Pt. Ltd. Here, I have done **250+ customized agitators**. I have **developed a marine propeller** for homogeneous mixing, which is **8.53% more efficient** than existing one. **Developed and implemented Impeller balancing** system that helps reduce power consumption by ~2 to ~5% and less vibration. I worked on product development of customizing agitators as per customer requirements & market demand. I also **designed many fixtures, jigs, Industrial equipment, and packaging solutions** for the big consignment. I also did **Project Management, precise weight calculation, cost calculation, and continuous improvement**. **Implemented ISO 9001:2015**– conducted internal Audits as Auditor.

Schneider Electric Infrastructure Limited, Halol, Vadodara

July 2017 – Dec 2017

Trainee Method Engineer (Full-time Internship 7th semester – part of curriculum)

I worked as a trainee method engineer in the Method department at the Inductor Manufacturing unit

- **Designed & implemented new fixtures and Jigs** in their plant as per requirement.
- **Solved the bending problem of the bus bar** at the 100-tonne hydraulic press **with help of ANSYS**.
- **Analysis of the pouring mould process with the temperature at different areas with respect to time.**
- The **project** was to develop an **automatic system for melt heating**.
- I developed their new plant's processes like **Kanban, OWS (SOP), 5S, MRB System, and Safety PPEs**.
- Worked on their production layout and made some changes for better productivity.
- Manage daily their documentation of design data.
- **Communicate with many vendors for Quotation, Manufacturing, and negotiation with vendors.**
- **Engineering Drawing Sheet reading & manageability.**
- **MTM (Method Time Measurement) analysis of all operations at the production line.**

Extra Curriculum

- Took a **remote project** from **Cummins Turbo Technology**, UK (Pune CTCI center) as product engineer **lead 20 people team** to develop turbocharger for **CNG Engine for SCANIA Truck**, I am **responsible** for **Product verification & validation** processes and documentations.
- Participated in various technical events & non-technical events etc.
- **Volunteer** at my college's **first tech fest in LITMUS'15**.
- **Event organizer** of a **junkyard** at **LITMUS'16**. (Teamwork, leadership, Management, financial approval)
- **Event organizer** of **BIG BOSS** at **LITMUS'17**. (Teamwork, leadership, managing people, financial approval) One-day side job as a salesperson in real estate opening at Golden Space Real Estate Pvt. Ltd.
- **Done Marketing** for the **"Exotic Valentine's Day" event at Vadodara**. It was a pleasant experience.

Engineering College Projects

1) POCKET WIND CHARGER At "ITMVU" international conference (1st Sem)

This project is on renewable energy. A small device produces electricity to charge the mobile phone during traveling.

2) Quad Wheel Trolley (Final Project – Last (8th) Semester)

It can carry a load of up to 150kg, which means almost all furniture and household objects can be carried taken less effort. It's used for climbing stairs with a load.

Achievements

- Runner-up up in -1) Robot War at ITMVU, 2) Robot Race at DAIICT
- Innovator of the month award
- **Utility Patent granted for Ground Mount Solar Structure (Manual Tracking)**

SKILLS

- Teamwork, Team Management, quick decision-making, Leadership.
- Strong willpower and we can do attitude.
- Creativity and innovation.

Hobbies

- Learning new technologies and teach others.
- Observe the nature and travelling.
- Watching Motivational Movies.
- Spending more time with my family.
- Cooking and New food try

Languages

- English, Hindi, Gujarati